PRODUCT REQUIREMENTS DOCUMENT (PRD)

Autonomous Version-Aware Document Intelligence Engine with Verifiable Grounding

Stage 2 Engineering Specification | Hackathon Problem Statement SH-SVA-04

Product Name	RAGNext (Cyber Knights Document Intelligence)
Document Type	Product Requirements Document (PRD) — Release v1.0
Problem Statement ID	SH-SVA-04 (Smart Virtual Assistant for Version-Aware Document Search)
Team Members	Sainath Loganathan, Tahir Azeem
Core Architecture	Temporal Version Lineage Graph, Pre-Retrieval State Filtering, Local Ollama Inference
Target Environment	Enterprise Air-Gapped / Offline-First / Zero Cloud Telemetry

Core Principle: *'Find the right information, from the right document version, with verifiable evidence.'*
RAGNext eliminates superseded document hallucinations by combining revision lineage graphs, metadata pre-retrieval filtering, and exact page-level grounded citations.

Problem First – Architecture Next – Verifiable Intelligence Always

Table of Contents

#	Section	#	Section
1	Executive Summary & Product Vision	6	Non-Functional Requirements (NFR)
2	Problem Statement & Scope	7	Data Model & Database Schema
3	User Personas & User Journeys	8	REST API Endpoints Specification
4	System Architecture & Tech Stack	9	UI & Screen Workflows
5	Functional Requirements (MoSCoW)	10	Success Metrics & Milestones

1. Executive Summary & Product Vision

1.1 Product Vision: RAGNext is an autonomous, version-aware enterprise document intelligence engine designed to eliminate superseded clause hallucinations. Unlike conventional vector-only RAG pipelines, RAGNext embeds temporal lifecycle states directly into document embeddings and metadata graphs, guaranteeing that employees and compliance officers receive authoritative, current-version information with verifiable page-level evidence.

1.2 Business Impact: 100% elimination of outdated policy citations, 60% reduction in document audit turnaround time, complete defense against regulatory non-compliance fines, and zero cloud API dependency through local air-gapped LLM inference.

2. Problem Statement & Scope Boundaries

Scope Area	In-Scope (Current Release v1.0)	Out-of-Scope (Future Releases)
Document Formats	PDF, DOCX, TXT, Markdown with page-aware extraction.	Scanned raster images, audio/video transcripts.
Version Lifecycle	DRAFT, ACTIVE, SUPERSEDED, ARCHIVED states with auto-transitions.	Git-style three-way file merging of conflicting clauses.
Retrieval Logic	Pre-retrieval temporal filtering, explicit historical intent routing.	Cross-lingual translation of foreign language policies.
Inference & Storage	Local Ollama (llama3.2 + nomic-embed-text), SQLite, local vector store.	Third-party cloud AI APIs (OpenAI/Anthropic) to maintain sovereignty.

3. User Personas & System Workflows

Persona Profile	Goals & Responsibilities	RAGNext Solution Feature
Rajesh Compliance & Risk Auditor	Audits historical policies; verifies clause validity at past incident dates; avoids legal penalties.	Historical Version Query Mode; automatic active conflict sentinel; downloadable audit logs.
Priya Knowledge Worker / Employee	Needs rapid, trusted answers to policy questions (remote work security, leave benefits).	Default Active Pre-Filtering; returns clear badges (v3.0 [Active]) with verbatim page citations.
Dev / IT Admin Systems Administrator	Ensures system availability; protects company data privacy; avoids high cloud API costs.	100% offline local inference via Ollama; zero external cloud data transfer.

4. System Architecture & Technical Specifications

RAGNext employs a decoupled, three-tier architecture ensuring high throughput, modularity, and zero cloud dependency:

Tier	Technology Component	Operational Function
Presentation Tier	React 18 + Vite + Bootstrap 5 + Lucide	Responsive dashboard, document library, version graph viewer, and AI chat with citation cards.
Application Tier	Node.js 18+ / Express.js (ES Modules)	REST API gateway, multipart document parser, temporal filter engine, and conflict detector.
Parsing Engine	pdf-parse, mammoth (DOCX), text/markdown	Page-aware coordinate extraction, 500-char semantic chunking with 100-char overlap.
Metadata Storage	SQLite / PostgreSQL (Dual Adapter)	Relational integrity for document groups, versions, statuses, and audit event logs.
Vector Store	Embedded Persistent Vector DB / ChromaDB	High-speed cosine distance calculation with pre-retrieval version ID filtering.
Inference Engine	Ollama (llama3.2 + nomic-embed-text)	Local 768-dim embeddings and conversational grounded synthesis with deterministic fallback.

5. Functional Requirements (MoSCoW Matrix)

ID	Priority	Functional Requirement	Detailed Acceptance Criteria
FR-01	MUST	Multi-Format Ingestion	Ingests PDF, DOCX, TXT up to 25 MB; generates SHA-256 checksum and preserves page boundaries.
FR-02	MUST	Document Grouping	Associates multiple versions into a unified parent group (e.g., 'Remote Work Security Policy').
FR-03	MUST	Lifecycle Management	Supports ACTIVE, SUPERSEDED, DRAFT, ARCHIVED states; transitions older versions when promoted.
FR-04	MUST	Context-Aware Chunking	Splits text into 500-char chunks with 100-char overlap; attaches page_number and version_id.
FR-05	MUST	Vector Embeddings	Embeds chunks via local Ollama (nomic-embed-text) with fallback token frequency vectors.
FR-06	MUST	Temporal Pre-Filter	Eliminates non-active versions from retrieval candidate set before cosine distance ranking.
FR-07	MUST	Historical Query Routing	Detects queries targeting historical revisions (e.g. 'in v1.0') and routes retrieval accordingly.
FR-08	MUST	Verifiable Citations	Returns document name, version tag, page number, and verbatim excerpt for every synthesized claim.
FR-09	MUST	Conflict Sentinel	Flags a prominent alert when multiple active policies coexist within the same group.
FR-10	MUST	Missing Evidence Gate	Returns explicit insufficient evidence status if similarity score < 0.45 or query is out-of-scope.
FR-11	SHOULD	1-Click Seed Dataset	Dashboard button instantly loads Leave Policy and Remote Security Policy test suites.
FR-12	SHOULD	Audit Telemetry Log	Records query string, retrieved chunk IDs, cited versions, latency, and timestamp to audit store.
FR-13	COULD	Side-by-Side Diff View	Interactive UI diff showing clause additions, removals, and changes between two revisions.

6. Non-Functional Requirements (NFR)

ID	Category	Target Metric	Validation Method
NFR-01	Inference Latency	< 3.5 seconds response time	Benchmarked on local Ollama llama3.2 using 50 consecutive queries.
NFR-02	Retrieval Speed	< 150 ms for 10,000 chunks	Vector distance timing tracked in backend processing middleware.
NFR-03	Data Sovereignty	100% Air-Gapped / Zero Cloud Outbound	Network packet monitoring verifying zero outbound requests during inference.
NFR-04	Fault Tolerance	100% Service Availability Fallback	Automatic switch to deterministic extractive synthesis if Ollama is halted.
NFR-05	Security & Sanitization	Cryptographic file hashes; no path traversal	Path sanitization tests preventing '../' injection in upload routes.

7. Data Model & Database Schema

RAGNext structures enterprise documents into an explicit relational hierarchy ensuring temporal integrity and cascade-safe updates:

Table Name	Key Columns	Relationships & Constraints
document_groups	id, name, category, description, created_at	Top-level grouping container; 1-to-N with document_versions.
document_versions	id, group_id, version_string, status, effective_date, file_path, checksum, is_conflict	Foreign Key ➔ document_groups(id) ON DELETE CASCADE. Enforces status lifecycle.
chunks	id, version_id, chunk_index, page_number, content, token_count, vector_id	Foreign Key ➔ document_versions(id) ON DELETE CASCADE. Stores page coordinates.
audit_logs	id, query_text, version_intent, retrieved_chunk_ids, cited_version_ids, latency_ms	Immutable query audit trail required by ISO 27001 and NIST 800-53 standards.

8. REST API Endpoints Specification

Method	Endpoint Route	Payload / Parameters	Response Description
GET	/api/health	None	Returns server, database adapter, and vector store health status.
GET	/api/documents	None	Lists all document groups, version counts, and active version details.
POST	/api/documents/upload	file (multipart), groupId, version, status	Parses document, extracts pages, chunks text, generates embeddings.
GET	/api/versions/:groupId	groupId (path)	Returns ordered revision lineage tree with conflict indicators.
PATCH	/api/versions/:id/status	{ status: 'ACTIVE' | 'SUPERSEDED' }	Updates version lifecycle state; propagates supersession.
POST	/api/chat	{ query, groupId, versionFilter }	Executes version-aware RAG, returns answer with exact citations.
POST	/api/demo/seed	None	Seeds multi-version Leave Policy and Remote Security Policy test files.

9. UI & Screen Workflows

Screen / Route	Key Visual Elements	User Actions Supported
Dashboard (/)	KPI cards (Docs, Active, Superseded), 1-Click Demo Seed, quick activity feed.	Quick status review, instantaneous demo dataset provisioning.
Document Library (/documents)	Grouped document cards, active version badges, format badges.	Inspect revision histories, download files, delete outdated groups.
Ingestion Portal (/upload)	Drag-and-drop file zone, metadata inputs, parsing progress bar.	Upload new document revisions, assign version numbers, set status.
AI Chat (/chat)	Conversational bubble stream, citation cards, conflict alert banner.	Ask natural language questions; inspect exact version, page, and excerpt.
Version History (/versions/:id)	Interactive timeline (v1.0 ➔ v2.0 ➔ v3.0), status toggles.	Promote draft to active, supersede versions, view clause diffs.
Settings (/settings)	Ollama connectivity pill, model selector, DB adapter indicator.	Test local LLM connection, inspect vector chunk counts.

10. Success Metrics, Test Matrix & Release Milestones

Key Performance Indicator (KPI)	Benchmark Target	Observed Test Result
Superseded Clause Citation Rate	0.0% (Zero Tolerance)	0.0% — Active filter successfully rejects v1.0 and v2.0 clauses.
Grounded Citation Precision	100% with Page Coordinates	100% — Every claim includes doc title, version, page, and quote.
Missing Evidence Rejection	0.0% Hallucinations on Unseen Data	Verified — Reports 'insufficient evidence' for out-of-scope queries.
End-to-End Local Inference Speed	< 4.0 seconds per query	1.2 - 2.8 seconds on Ollama llama3.2 (local host).
Offline Air-Gap Compliance	100% Zero External Requests	Verified — Zero outbound telemetry or external cloud API calls.

Release Milestones & Delivery Roadmap:

Phase	Milestone Deliverables	Status
Stage 1 (Framing)	Problem definition, root-cause analysis, market opportunity framing PPT.	**COMPLETED**
Stage 2 (MRD Submission)	Formal 9-page Market Requirements Document (PDF) with user personas and business model.	**COMPLETED**
Stage 2 (PRD & MVP Release)	Functional React 18 UI, Express backend, Ollama integration, automated RAG test suite, PRD.	**ACTIVE / OPERATIONAL**
Stage 3 (Enterprise Scale)	OCR scanner integration, enterprise SharePoint connectors, Active Directory RBAC.	PLANNED ROADMAP